Prompt-Driven Interactive Segmentation

as a Dataset-Annotation Tool

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Abstract

Producing accurate instance masks for image datasets is time-consuming when every object must be outlined manually. We propose a local interactive annotation tool that uses positive/negative point and box prompts with the frozen SAM 2.1 Hiera-Tiny model, and measures annotation time and mask agreement with existing ground truth. We will use a deterministic 300-image subset of COCO 2017 val2017, retain its non-crowd instance masks, and compare prompted annotation against manual polygon labeling on the same target objects. The project will report per-instance intersection over union (IoU), Dice, annotation time, and inference latency without training a model or claiming results before they are measured.

1 Problem Statement & Motivation

Instance segmentation assigns a pixel mask to each object in an image, but creating these masks by hand is labor-intensive and can limit the scale of downstream image-processing experiments. This project investigates whether prompt-driven segmentation can reduce human annotation effort while retaining useful agreement with reference masks. The work connects directly to the course topics of digital image segmentation, region delineation, binary-mask representation, and quantitative evaluation. Rather than train a new segmentation model, we will build a reproducible human-in-the-loop tool around a frozen pretrained model and evaluate both the resulting masks and the time required to accept them.

2 Related Work

Promptable segmentation builds on interactive segmentation and recent foundation-model approaches. GrabCut established a classic interactive foreground-extraction method in which user input guides iterative graph-cut optimization (Rother et al., 2004). The Segment Anything Model (SAM) introduced a promptable model trained at large scale and evaluated with point/box prompts (Kirillov et al., 2023). SAM 2 extends promptable segmentation to images and videos with a streaming memory architecture; we use its image predictor rather than training a model from scratch (Ravi et al., 2024). FastSAM is a recent alternative that seeks faster segment-anything behavior with a different model pipeline (Zhao et al., 2023).

Our project is a small systems and evaluation study, not a new foundation model or a reproduction of SAM 2 training. COCO provides the images and instance masks needed for a post-hoc

comparison (Lin et al., 2014). We will compare an interactive SAM 2.1 workflow with manual polygon annotation on matching target objects. The evaluation will separate model inference latency from end-to-end annotation time, and will use the held-out COCO mask only for scoring, not to choose prompts or candidate masks.

Table 1: Key references.

Reference	Year	Relevance to this project
(Rother et al., 2004)	2004	Classic interactive foreground extraction guided by user input.
(Lin et al., 2014)	2014	Public images and instance-segmentation ground truth for evaluation.
(Kirillov et al., 2023)	2023	Promptable segmentation with points, boxes, and masks.
(Zhao et al., 2023)	2023	Recent promptable segmentation alternative for related-work context.
(Ravi et al., 2024)	2024	Frozen image/video segmentation model used by our tool.

3 Dataset & Baseline Plan

We will sample 300 images from COCO 2017 val2017. Candidate images must have between one and eight non-crowd instance annotations; candidates are sorted and sampled with Python's pseudorandom generator using seed 42. We will record the selected image IDs, retain the corresponding instance annotations, and exclude crowd regions from scoring. The COCO images retain their individual source licenses, so we will follow the license metadata for every image and will not redistribute the images or derived masks. COCO's terms are documented by the dataset maintainers (COCO Consortium).

The primary baseline is manual polygon annotation of the same target object in each evaluated image, using a standard polygon annotation interface without SAM assistance. Student A will record active annotation time using the same start and stop definition used by the interactive tool. For the prompted method, Student B will record time from the first prompt through accepting the mask; inference latency will be logged separately. Before either method is run, the target COCO instance ID will be recorded. After each mask is finalized, we will compare it with its matching COCO non-crowd instance mask. The reference mask will not be exposed while prompts are chosen.

Table 2: Dataset summary.

Dataset	Subset used	Link / terms
COCO 2017	300 val2017 images; seed 42; 1–8 non-crowd instances per image	https://cocodataset.org/

4 Proposed Method

The application has a React/TypeScript browser interface and a Python/FastAPI backend. The backend runs the frozen SAM 2.1 Hiera-Tiny image predictor with positive or negative point prompts and/or a box prompt. For each request, the predictor returns candidate masks and confidence scores; the tool displays the highest-scoring candidate for human review. The annotator can refine prompts, accept the mask, and save it. The evaluation pipeline associates the accepted mask with its COCO annotation ID, then computes binary-mask IoU and Dice against the corresponding ground truth. The annotation timer starts at the first prompt and ends when the annotator accepts the mask; inference time is measured separately. The main contribution is a small, reproducible annotation-and-measurement workflow, not a change to SAM 2's model architecture.

5 Feasibility & Compute Plan

The project is CPU-OK because it performs inference with a frozen, small SAM 2.1 checkpoint and does not train or fine-tune the model. The selected Hiera-Tiny checkpoint is approximately 149 MiB; inference latency will be measured on the available machine and reported rather than assumed. The 300-image subset is small enough for local storage and evaluation. If CPU inference is too slow for the full study, we will first complete the MVP on a fixed smaller sample and report the actual sample size and device.

5.1 Minimum Viable Deliverable (MVP)

Run the interactive tool end-to-end on a fixed sample of 30 COCO images; measure manual polygon and prompted annotation time on matching target objects; save both methods' masks; and report mean/median annotation time, mean IoU and Dice, and per-image results. The sample IDs, prompts, device, model checkpoint, and timing protocol will be recorded so the comparison can be reproduced.

5.2 Stretch Goal

Expand the comparison to all 300 images, report results by object category and prompt type (click versus box), include representative failure cases, and run a small usability test with a second annotator.

6 Risk Assessment

- **CPU latency or memory limits:** use the Hiera-Tiny checkpoint, measure inference separately from annotation time, and reduce the evaluated sample to the predeclared MVP if needed; no training is required.
- **Dataset retrieval or license variation:** download only the selected COCO images from the official source, preserve the selected IDs and annotation metadata, and follow each image's source license; do not commit or publicly redistribute dataset assets.
- **Prompt and ground-truth mismatch:** identify the target instance before annotation, keep its COCO annotation ID with the result, and score only after acceptance; crowd regions and empty masks are excluded.

7 Work Breakdown

Table 3: Milestone task allocation. Student B combines the catalog's model and evaluation roles.

Milestone	Student A — Data & Baseline	Student B — Model + Evaluation
M1 (Wk 3)	Prepare the COCO subset and ground-truth masks; define the manual-polygon timing protocol.	Build the click/box interface, run SAM 2 inference, and define the IoU/time evaluation protocol.
M2 (Wk 6)	Measure manual polygon time on the shared sample and verify target-mask IDs.	Deliver the interactive tool and first IoU/time comparison; combine catalog roles B and C.
M3 (Wk 9)	Complete manual-baseline timing and data checks.	Polish the tool; complete the full comparison, failure analysis, and small user test.

AI Tool Usage Disclosure

AI assistance was used for initial code scaffolding and drafting/editing this proposal. The team will verify the implementation, citations, measurements, and final technical claims, and will revise the report to reflect work they have personally reviewed and understand.

References

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